

Reaction-Wheel Mounting Bracket: Launch-Load Finite Element Analysis

Tulika Chaudhury

July 31, 2026

Overview

Each of the satellite's three reaction wheels is mounted in an individual bracket – an angled plate braced by two gussets, bolted through a base flange, that holds the Maxon ECX FLAT 22 L motor at the wheel's spin-axis tilt. The three wheels are mutually orthogonal, each inclined at $\cos^{-1}(1/\sqrt{3}) \approx 54.74^\circ$ from the Z axis and spaced 120° apart in azimuth. The motor-seating face, measured directly from the CAD model, is inclined 35.26° from the base plane – the exact complement of 54.74° , confirming that the bracket geometry is consistent with the intended wheel configuration.

This report presents a launch-load finite element analysis (FEA) of the bracket, performed on the as-uploaded CAD geometry rather than a simplified representation. The objective is to determine whether the bracket, as designed, survives the structural and vibrational loading of launch. Small, light, cantilevered metal components of this type are particularly susceptible to cracking or resonance-driven failure if under-designed.

Load Case

Launch loads for a structure of this size are conventionally represented by a *quasi-static* analysis: rather than simulating the full random-vibration environment, the structure is analyzed under a steady, elevated gravitational load. A value of 12g is the standard conservative figure used in the absence of a specific launch-vehicle specification, intended as a worst-case equivalent for several seconds of vibration rather than a literal instantaneous load.

The 12g load was applied separately along each of the X, Y, and Z axes, since a bracket that performs well under one loading direction may not perform as well under another. A safety factor of 1.25 was applied on top of the 12g case when evaluating margin, so the reported margins already incorporate this conservatism.

The applied load includes the bracket's own mass together with the motor and flywheel assembly bolted to it: 32 g for the Maxon motor (per its published datasheet) and 31.116 g for the flywheel, for a combined 63.116 g. This mass was modeled as a rigid lumped mass attached to the four motor screw holes, offset approximately 10 mm from the mounting face to approximate the combined motor-and-flywheel center of gravity.

Finite Element Model

The bracket was meshed with 24,332 nodes and 14,167 quadratic tetrahedral elements, refined to 0.35 mm in the vicinity of the bolt and screw holes, where stress concentration is expected. Every element satisfied a positive-Jacobian check prior to solving, confirming the absence of inverted or degenerate elements. Figure 1 shows the mesh on the deformed geometry.

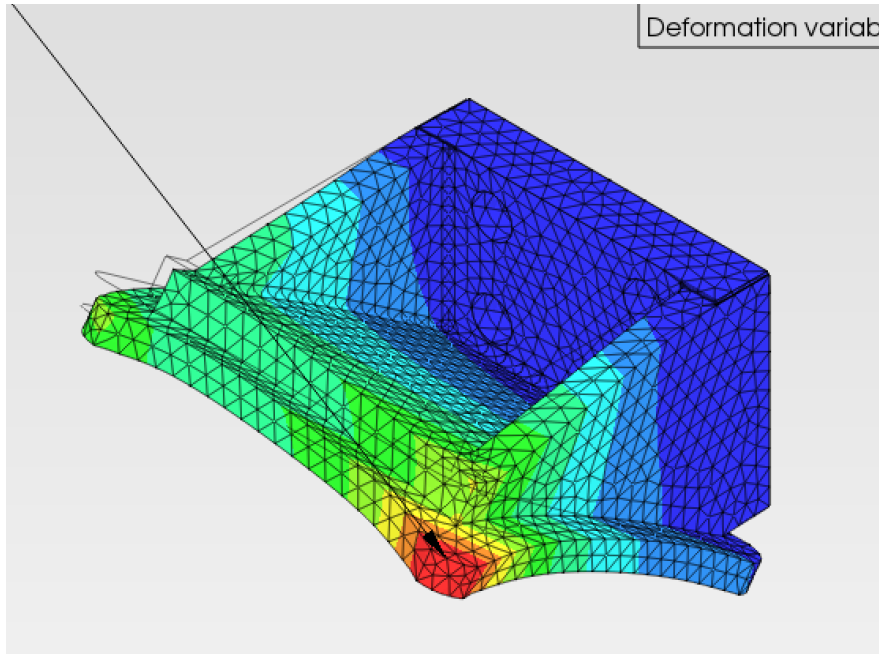


Figure 1: Bracket mesh on the deformed shape (displacement scaled for visibility).

Aluminium 7075-T6 ($E = 71.7$ GPa, yield strength 503 MPa) was used as the primary material, given the stress-concentration features present (gusseted corner, small bolt holes). Aluminium 6061-T6 (yield strength 276 MPa) was also evaluated for comparison; the resulting stress field is materially unchanged between the two alloys, since stress in this analysis is governed primarily by geometry and loading rather than material selection, making this comparison effectively a secondary yield-strength check rather than an independent analysis.

The base flange – the four bolt holes through which the bracket is secured to the central mounting plate – was modeled as fully fixed, representing a properly torqued bolted joint.

Results

Table 1: Peak stress, deflection, and margin of safety, by load direction.

Load case	Peak stress (raw)	Peak stress (field) ¹	Max deflection	MS, 7075-T6	MS, 6061-T6
12g, +X	2.21 MPa	1.92 MPa	0.19 μm	+181	+99
12g, +Y	4.15 MPa	3.49 MPa	0.35 μm	+96	+52
12g, +Z	4.59 MPa	3.94 MPa	0.34 μm	+87	+47

Margin of safety is defined as $MS = S_y / (1.25 \times \sigma_{vm}) - 1$. A margin of, for example, +150 indicates the part can withstand approximately 150 times the current stress before yielding. The margins reported here are large: the applied loads amount to only a few milli-newtons of force distributed across several millimeters of aluminum, which does not produce significant stress – a physical outcome, not an artifact of the analysis. Figure 2 shows the location of peak stress (the base flange corner nearest the bolt holes, consistent with expectations for a clamped bracket foot under load), and Figures 3–4 present the same results graphically.

Bracket FEA - 12g quasi-static, +Z

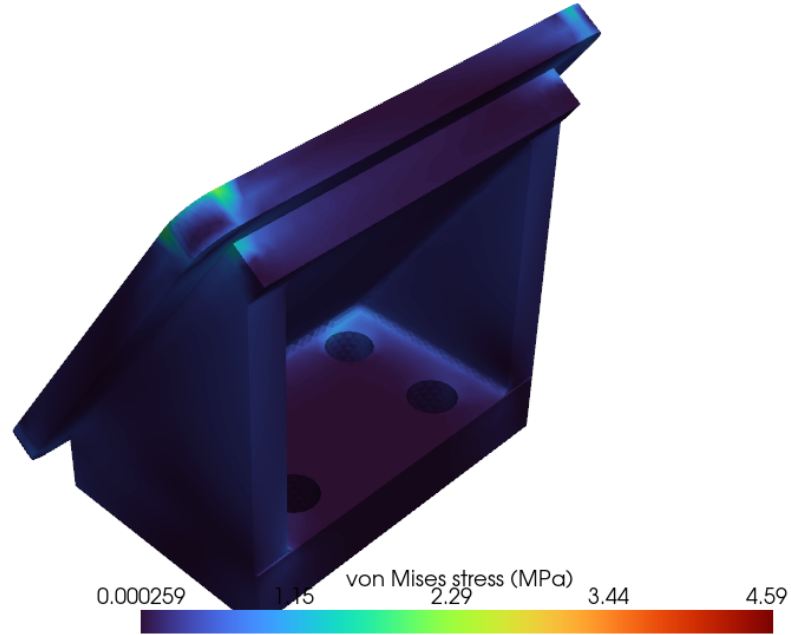


Figure 2: Von Mises stress contour for the worst case, 12g along +Z.

¹The raw peak-stress column includes a node located at a fixed-corner/bolt-hole boundary. Finite element models characteristically report an artificially high, mesh-dependent stress at such a boundary singularity, which does not represent a physical material limit. The field column excludes these boundary nodes and represents the more physically representative value. Margins of safety in this table are computed from the raw peak stress, the more conservative of the two readings; the overall conclusion is unaffected either way.

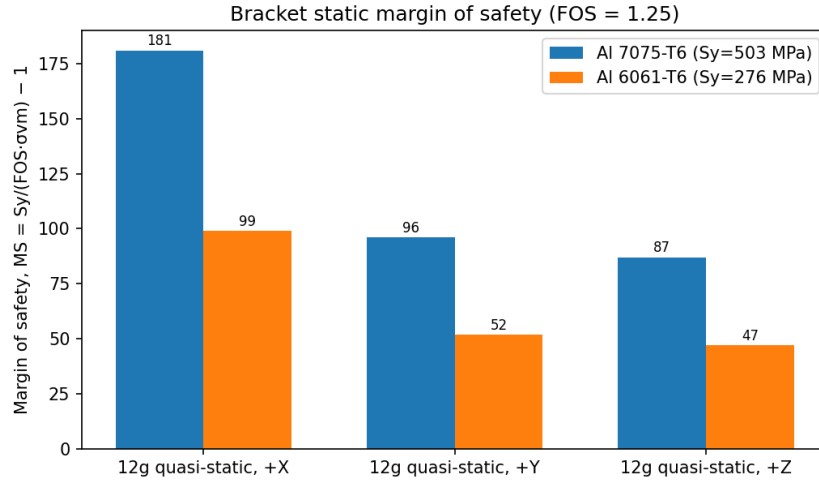


Figure 3: Margin of safety by load direction and candidate alloy.

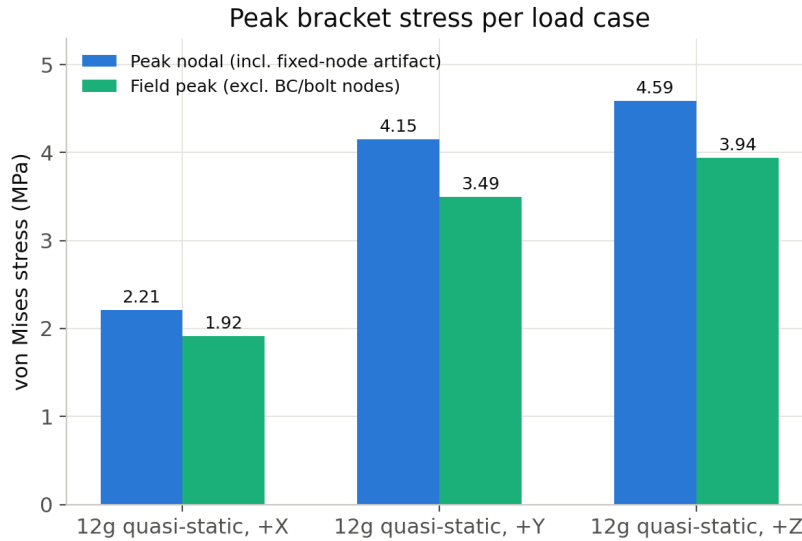


Figure 4: Peak stress per case: the raw (boundary-affected) value alongside the more representative field value.

Natural Frequency

With stress margins of this magnitude, stress is not the governing consideration. The parameter that determines whether a small, stiff bracket of this type survives launch is its *natural frequency* – its tendency to vibrate under its own dynamics. Launch vehicles impose vibration across a broad frequency spectrum, and a structure with an insufficiently high natural frequency risks resonance, which can produce substantially greater damage than a static load estimate would suggest, even where stress results appear favorable.

Using the same fixed-base model with the motor and flywheel mass attached, the first natural

frequency is

$$f_1 \approx 3253 \text{ Hz},$$

approximately 30 times the roughly 100 Hz minimum stiffness typically specified by CubeSat deployer requirements. The bracket is not expected to be a resonance risk during launch.

Limitations and Scope

Two points should be noted, as they affect the interpretation of these results.

The 12g load case represents a quasi-static approximation rather than a full random-vibration analysis. It is the standard first-order screening approach for a structure of this size, and reflects the conventional meaning of *launch loads* in this context; however, a complete random-vibration (PSD) analysis remains the appropriate next step prior to qualifying the bracket for launch.

The highest raw stress value occurs at a mathematically singular point – a perfectly clamped sharp corner – where any finite element model will report an artificially elevated, mesh-dependent stress that does not correspond to a physical material limit. Table 1 accordingly reports both the raw value and a field value that excludes the boundary nodes responsible for this artifact; the conclusion is unchanged in either case.

Conclusion

The bracket satisfies launch quasi-static loading requirements with substantial margin in every direction evaluated, and its first natural frequency is well clear of typical CubeSat stiffness requirements. The design does not present a structural risk; if anything, the results suggest room in a future iteration to reduce material for mass savings, as the part is over-designed relative to the demonstrated loads.